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WHO'S THAT METAL! A Learning-Curve Study using Infrared Spectra for Machine Learning Identification of Lanthanide Complexes Modeled with Transition Metals

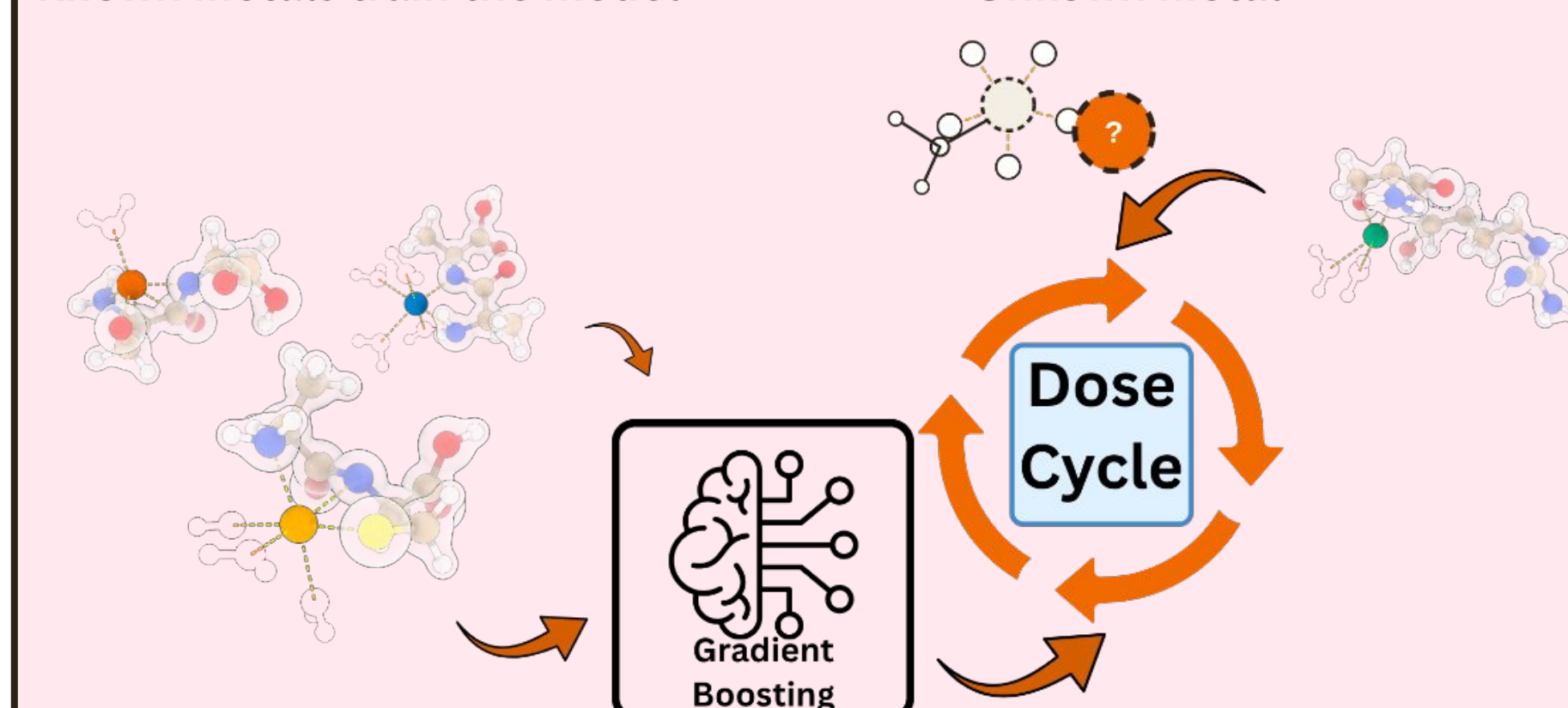
WHY IR?

Every molecule has a unique signature that is given off through the infrared spectra. Through using the signature, this project aims to see if it's significant enough to construct a pattern.

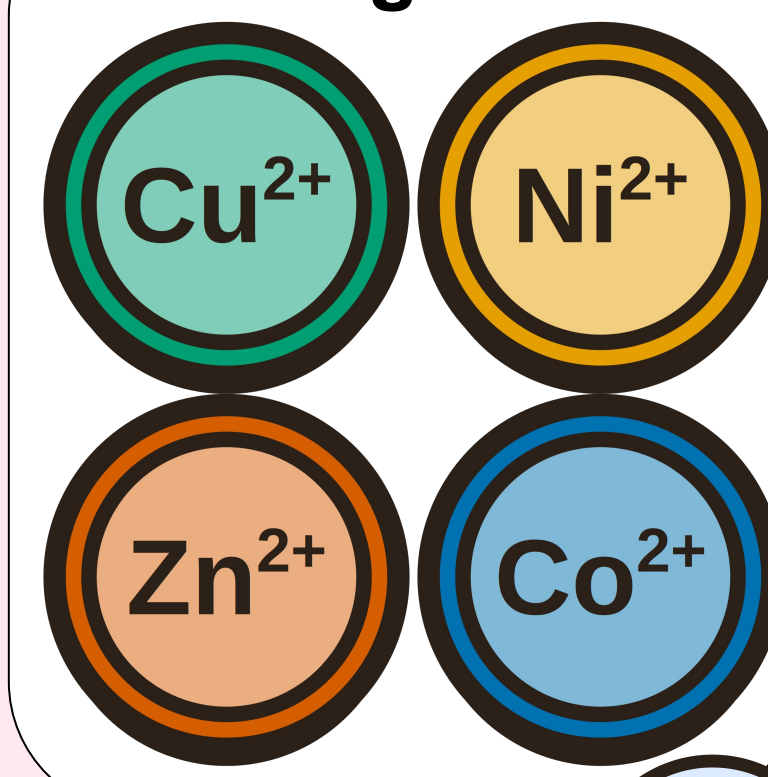
The Problem

This project aims to develop a pipeline that includes a machine-learning approach to classify metals attached to **peptide complexes**. This project centers on building the data pipeline that makes such modeling possible and on the **machine learning-curve** study that follows from it.

Known Metals train the model



Legend



THE DATASET

10,900 spectra

4 metals · ~2,700 each · balanced
791 bins per spectrum
(50–4000 cm^{-1} , steps of 5 cm^{-1})
Kept only when no frequency/intensity mismatch

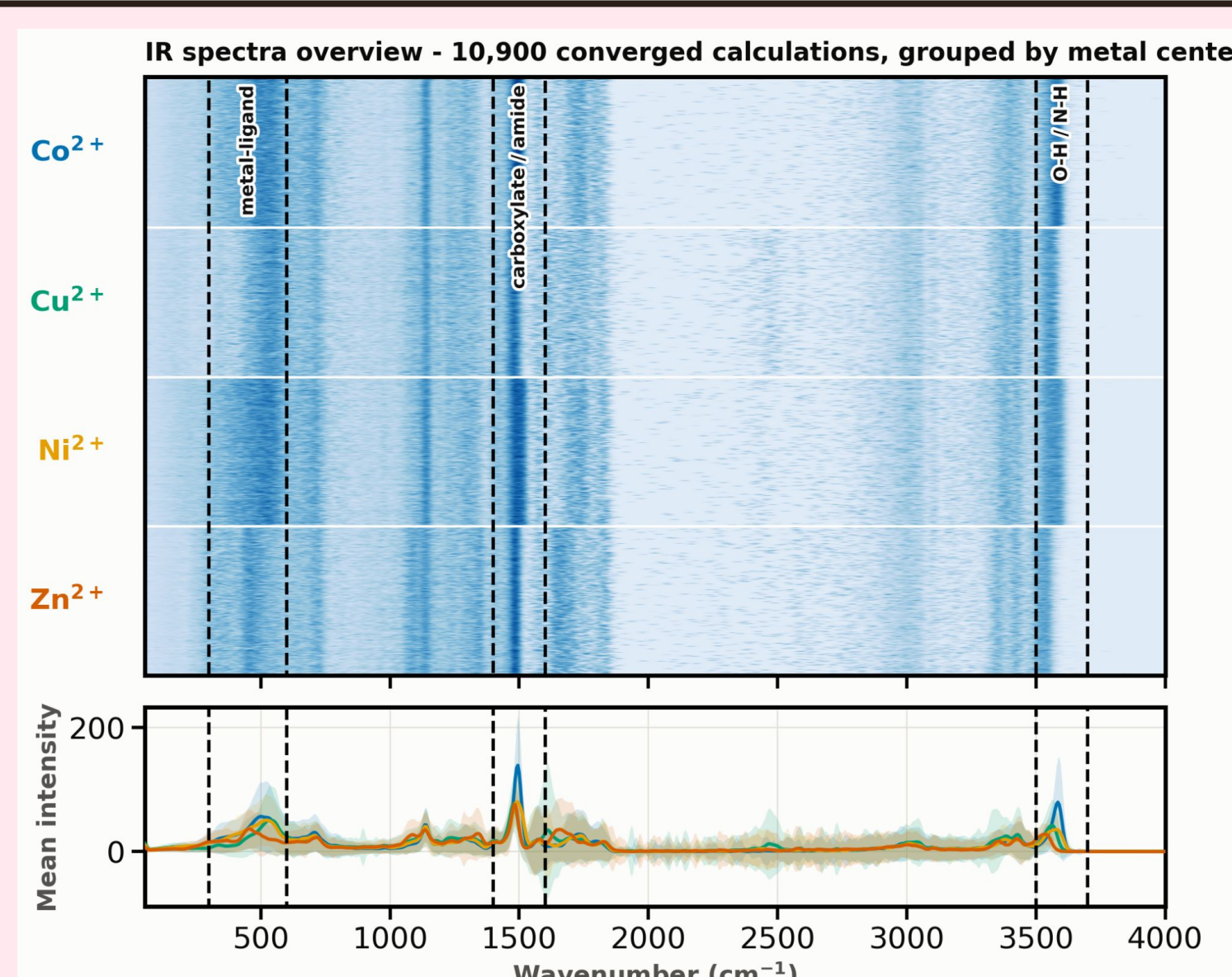
#	mode	symmetry	wavenumber	IR intensity
1	a	-0.00	0.00000	-
2	a	-0.00	0.00000	-
3	a	0.00	0.00000	-
4	a	0.00	0.00000	-
5	a	0.00	0.00000	-
6	a	0.00	0.00000	-
7	a	25.72	1.13128	YES
8	a	49.57	0.56819	YES
9	a	65.20	0.41024	YES
10	a	75.76	9.81047	YES
11	a	84.25	5.22847	YES
12	a	93.90	3.83901	YES
13	a	95.12	3.33275	YES
14	a	105.02	2.05246	YES
15	a	114.53	3.86610	YES
16	a	118.47	1.61204	YES
17	a	133.30	11.07456	YES
18	a	145.03	2.90593	YES
19	a	151.21	2.84784	YES

Data Exploration

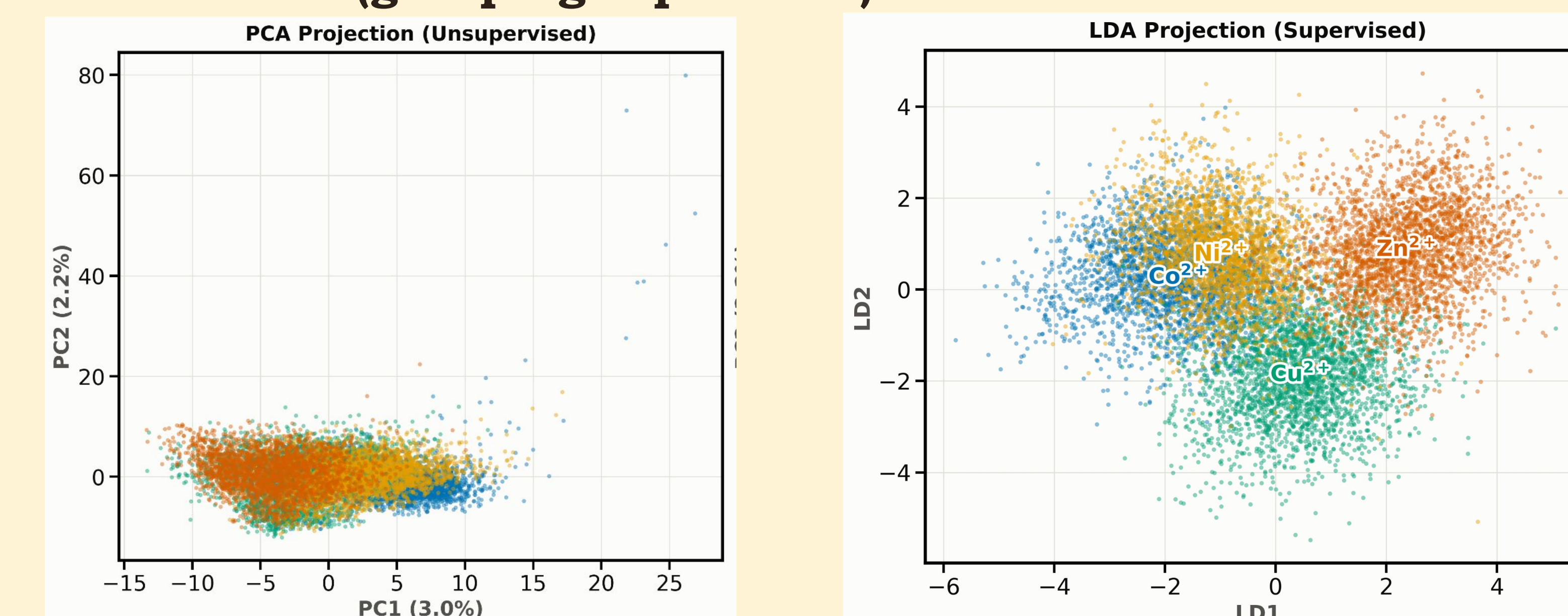
Feature importance concentrates in 3 coordination-sensitive bands:

300–600 cm^{-1} Metal–ligand stretch
~1500 cm^{-1} Carboxylate / amide
~3600 cm^{-1} O–H / N–H stretch

These regions showed the highest activity between variation in Amino Acid Complexes.



PCA VS. LDA (grouping separation)

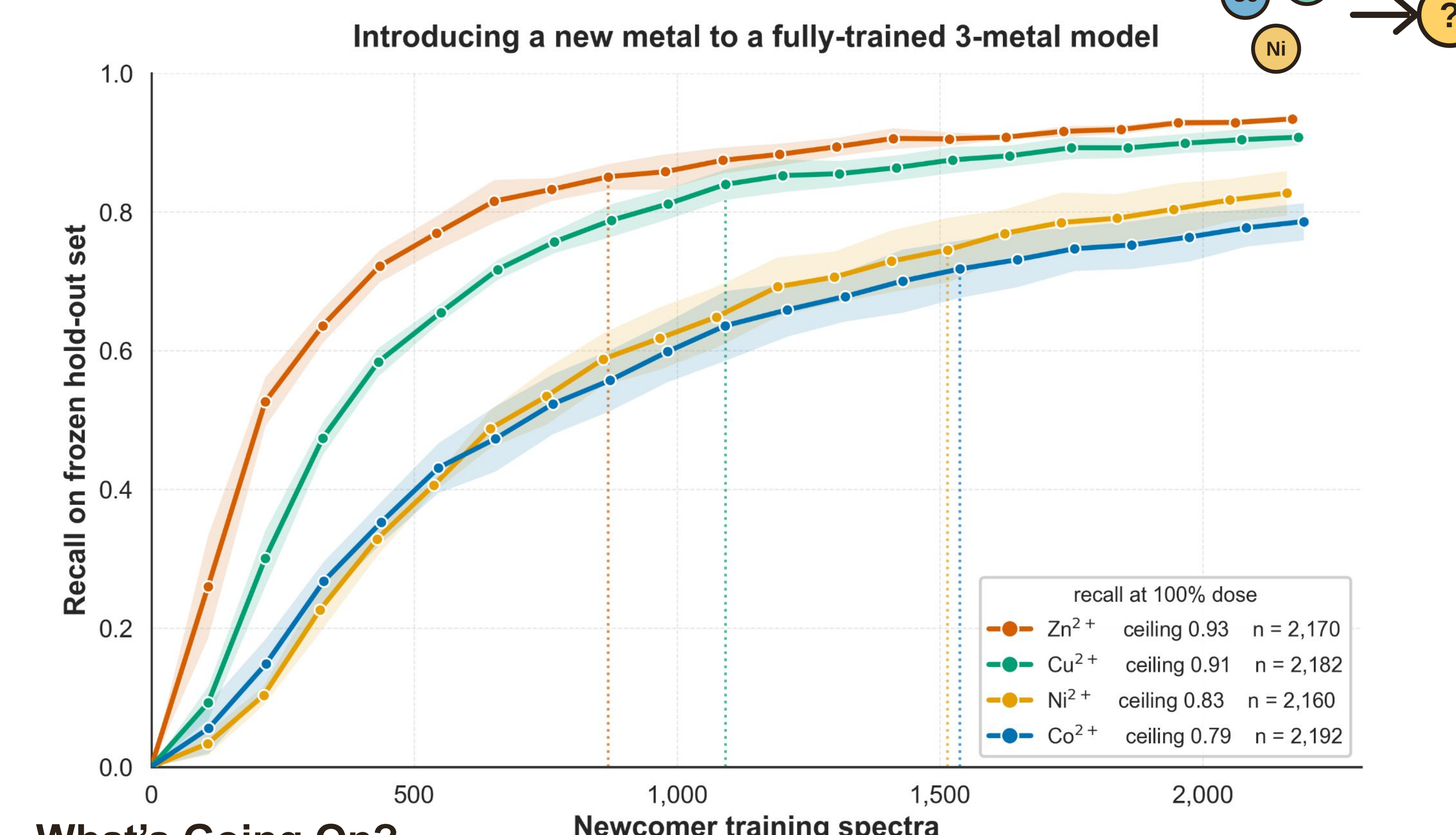


What's the Difference?

PCA (unsupervised) — metals don't separate along the top variance axes. The biggest variance is amino acid and conformation; metal identity is a *low-variance* feature, so an unsupervised projection can't identify it.

LDA (supervised) — a linear boundary does find the metal, proving the signal is real but plateaus giving further need for investigation in methods. We see some overlap clustering between Co and Ni hinting that it will be difficult to separate the two metals at high accuracy.

MEET A NEW METAL



What's Going On?

Each curve shows one metal introduced as a newcomer to a **Histogram Gradient Boosting** classifier already trained on the other three metals at full strength. Moving right, the newcomer's training data is added in 5% increments of its available pool, with the model retrained from scratch at each step and re-evaluated on a frozen hold-out set of that metal. The y-axis is recall on that hold-out set; shaded bands are the standard deviation across five independent hold-out splits. The dotted line highlights the region where we start to plateau in learning.

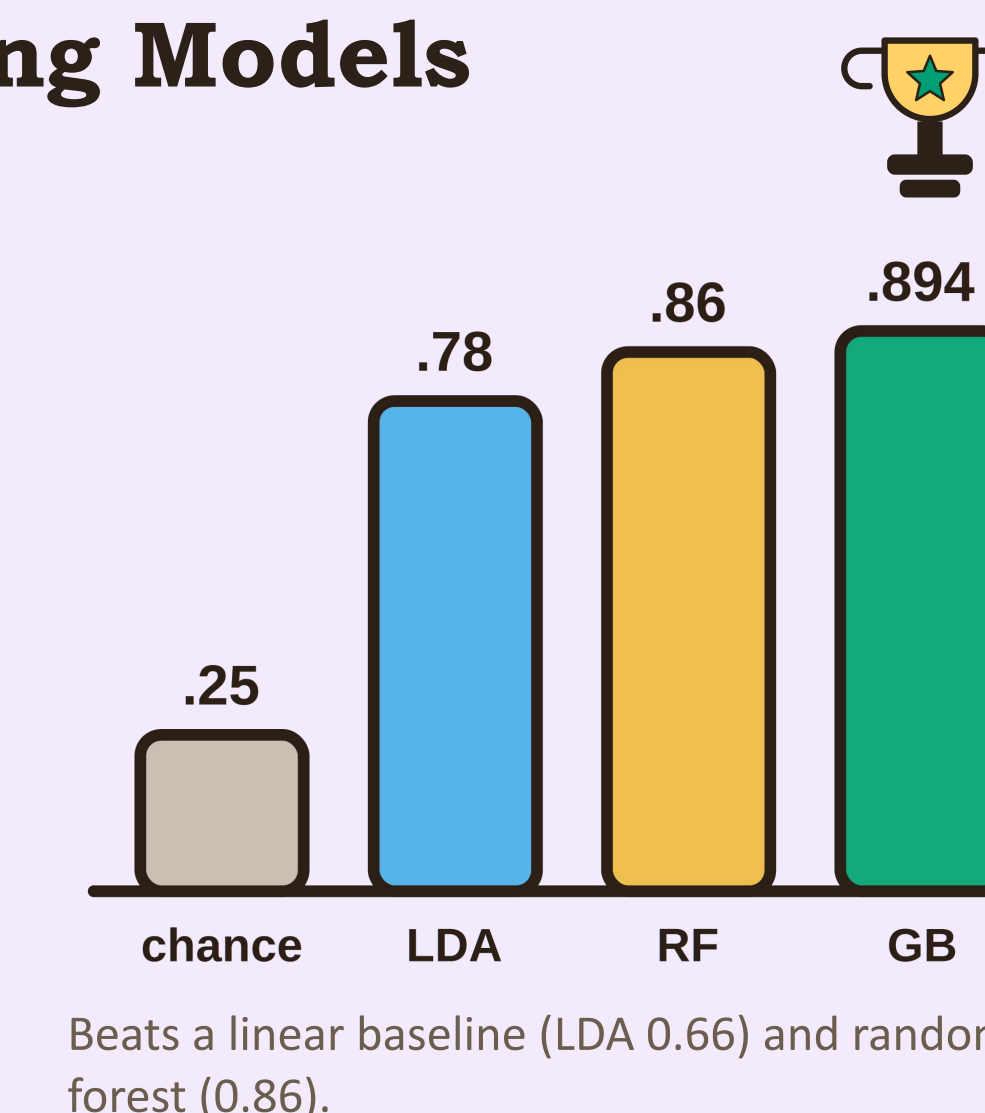
Difference in Machine Learning Models

Hist Gradient-Boosting (GB) Builds decision trees in sequence, each tree correcting the errors of the last.

Random Forest (RF) Averages many independent decision trees, each trained on a random part of the data and its features.

Linear Discriminant Analysis (LDA) Finds straight line boundaries that best separates the metal classes, taught us the signal was real but not linear.

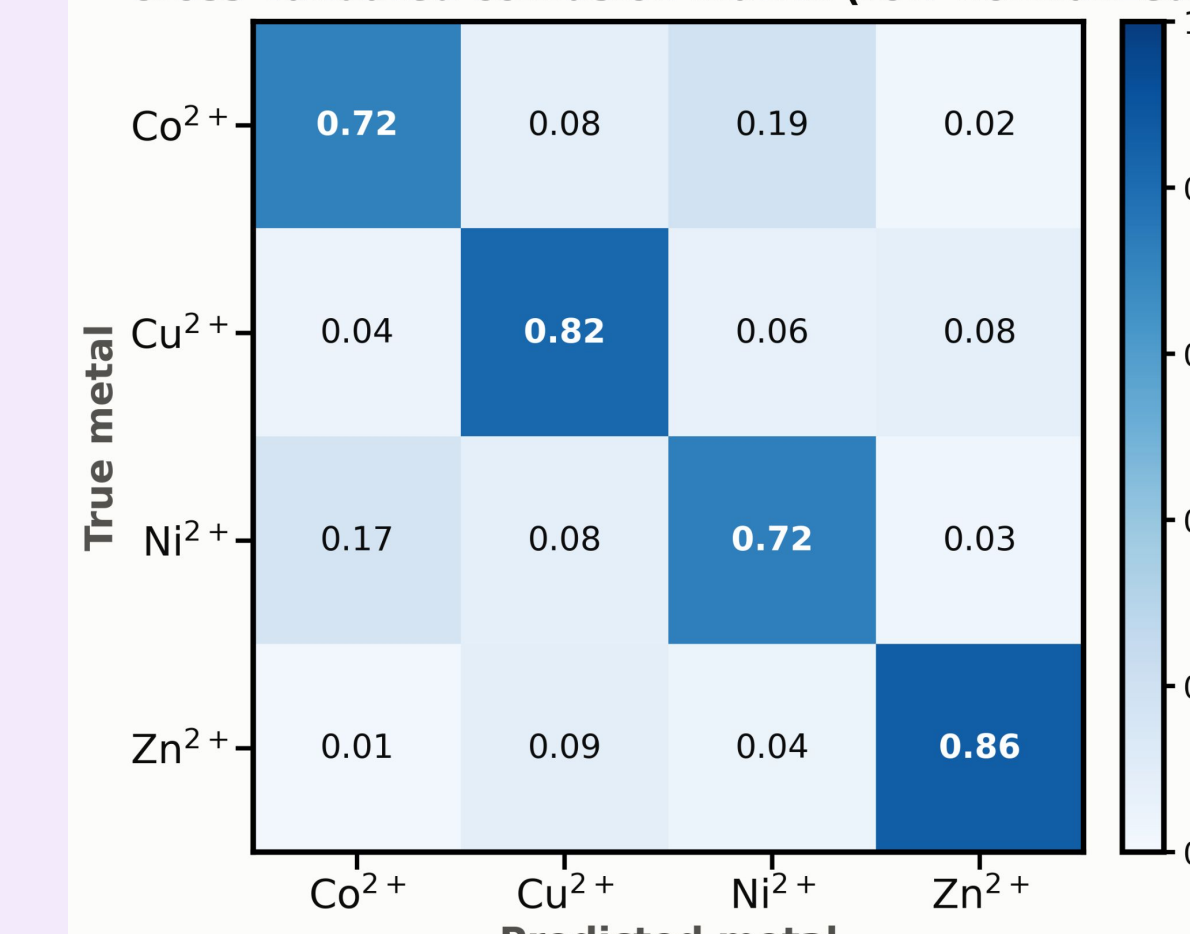
Chance Model is the accuracy you would expect if you were guessing at random, if a model beats this number it shows it has learned something



Cross Validation with Models

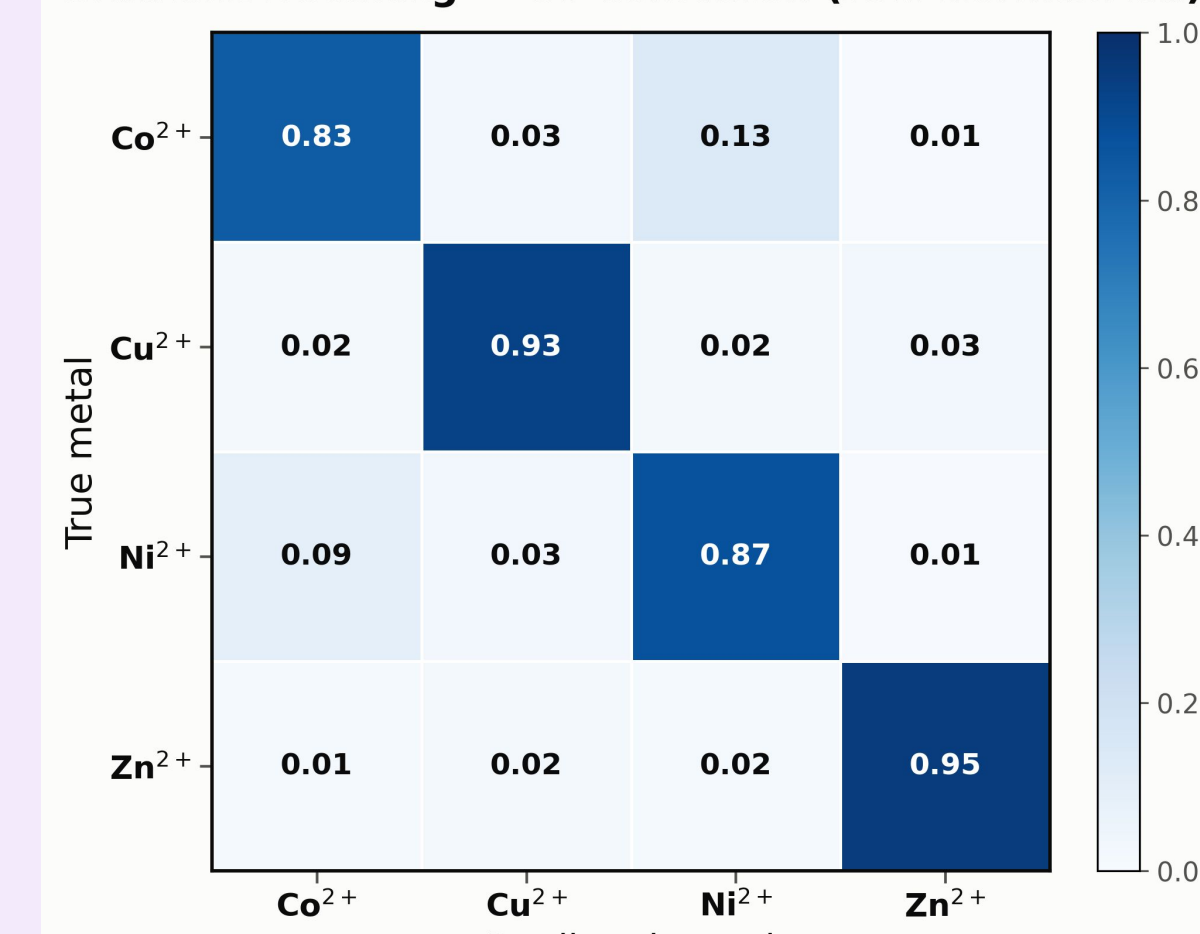
LDA ~ 78%

Cross-validated confusion matrix (row-normalized)



Gradient Boost ~ 90%

Gradient Boosting — CV confusion (row-normalized)



Significance and What Follows

Significance. The core question of this research was whether a metal's identity is recoverable from its IR spectrum, whilst the signal is buried under amino-acid variations - it is. Gradient Boosting identifies at 0.89 balanced accuracy against the chance model of 0.25. This sets a real basis for further exploration with experimental data.

What follows. The open question is whether this theory holds true to experimental IR spectra, not just xTB-Computed ones, that is where the real test lies. The newcomer-metal analysis shows positive results for a data budgeted approach to training the model for the pattern recognition in respect to predicting metal complexes.

Acknowledgements

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